A Complete and Natural Rule Set for Multi-Qudit Clifford Circuits in All Odd Prime Dimensions

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This talk

- A presentation by generators and relations of the n-qudit Clifford group, for every odd prime dimension p: 16 relations, none involving more than 3 qudits.
- A unique normal form for symplectic (= Clifford modulo Pauli) circuits.
- All, except for two math facts, are formalised in Agda.

About this slides

- Every Agda code block is typechecked by Agda.
- Written by *Claude* and Xiaoning Bian ($\sim$200-line outline prompt and slides checking $\&$ improvement).

Introduction

From qubit to qudit Clifford gates

Fix an odd prime p and let $\omega = e^{2\pi i/p}$. The Clifford group $\mathcal{C}_n$ is generated by:

	qubit ($p = 2$)	qudit (odd prime p)
H	$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$	$\lvert j\rangle \mapsto \frac{1}{\sqrt{p}} \sum_k \omega^{jk} \lvert k\rangle$
S	$\begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$	$\lvert j\rangle \mapsto \omega^{j(j-1)/2} \lvert j\rangle$
CZ	$\mathbf{diag}(1, 1, 1, -1)$	$\lvert j\rangle\lvert l\rangle \mapsto \omega^{jl} \lvert j\rangle\lvert l\rangle$

Note the $/2$ in the qudit S is the inverse of 2 in $\mathbb{Z}_p$. A global phase can be added to the definition of H to facilitate computation.

The setting: one Agda module

These slides are a single Agda module, parameterised by the prime p and by a primitive root g of $\mathbb{Z}_p^*$:

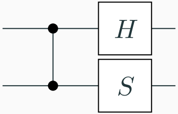
```
module qpl26slides.talk
  (p-3 : ℕ)
  (let p-2 = 1+ p-3)
  (p-prime : Prime (3+ p-3))
  (let open PrimeModulus' p-2 p-prime)
  (g*@(g , g≠0) : ℤ* ₚ)
  (g-gen : ∀ ((x , _) : ℤ* ₚ) → ∃ \ (k : ℤ ₚ₋₁) → x ≡ g ^′ toℕ k)
  where
```

- $p =$ p-3 $+ 3$: every odd prime, uniformly.
- Everything downstream — rule sets, normal forms, theorems — is parameterised by the choice of g.

Circuits are words

A gate set is a family of gate types indexed by arity; a circuit on n wires is a word over wire-indexed generators.

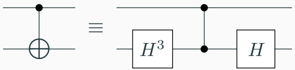
```
data SympGate : ℕ → Set where
  H-gate  : SympGate 1
  S-gate  : SympGate 1
  CZ-gate : SympGate 2

example : Circuit 2
example = CZ • H ↑ • S
```

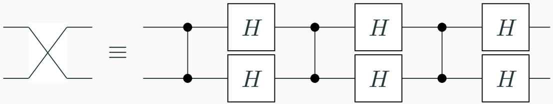
- `Circuit` n $=$ `Word` (`Gen` n): free monoid; `_↑` shifts a circuit up one wire.
- diagrams read in matrix-multiplication order: `example` $=$ 

Derived gates: the rest of the usual gate set

X, Z, the swap Ex, and the two controlled-X gates are *words*, not extra generators:

```
Z X : ∀ {n} → Word (Gen (₁₊ n))
Z = H • H • S • H • H • S⁻¹
X = H • S • H • H • S⁻¹ • H

Ex CX XC : ∀ {n} → Word (Gen (₂₊ n))
Ex = CZ • H ↓ • H ↑ • CZ • H ↓ • H ↑ • CZ • H ↓ • H ↑
CX = H ↓ ^ 3 • CZ • H ↓
XC = H ↑ ^ 3 • CZ • H ↑
```





Main results

Main result 1: a presentation of the qudit Clifford group

The paper's Figure 1 rule set, mod scalars, presents the projective Clifford group $\mathcal{C}_n/\langle\omega\rangle \;\cong\; \mathbf{Pauli}_n \rtimes \mathbf{Sp}(2n, \mathbb{Z}_p)$:

```
Theorem-1 : ∀ (n : ℕ) → (n CRel,_===_) IsPresentationOf (Pauli⋊Sp n)
Theorem-1 = PapPres.presentation
```

- `_CRel,_===_` is the 16-relation rule set, shown on the next slides.
- The theorem holds *at every width* n, for *every* odd prime p and every primitive root g — the whole deck is parameterised by them.

The 16 relations, in Agda (one qudit)

```
data _CRel,_===_ : (n : ℕ) → WRel (Gen n) where
  order-S      :       (₁₊ n) CRel, S ^ p === ε
  order-H      :       (₁₊ n) CRel, H ^ 2 === M₋₁
  M-power      : ∀ k → (₁₊ n) CRel, Mg^ k === M (g^ k)
  semi-MR      :       (₁₊ n) CRel, R • Mg === Mg • R^ (g * g)
  order-SH     :       (₁₊ n) CRel, (S • H) ^ 3 === ε
  comm-HHSHHS :       (₁₊ n) CRel, H • H • S • H • H • S === S • H • H • S • H • H
```

- M_x is the multiplier $|j\rangle \mapsto |xj\rangle$, spelled as the word $R^{x^{-1}}\, H\, R^x\, H\, R^{x^{-1}}\, H$ with $R = S \cdot Z^{1/2}$.
- One-qudit relations talk about the powers of a fixed primitive root g — this is where odd primes differ from the qubit case.

The 16 relations, in Agda (two qudits)

```
order-CZ   : (₂₊ n) CRel, CZ ^ p === ε
order-Ex   : (₂₊ n) CRel, Ex ^ 2 === ε
comm-CZ-S↑ : (₂₊ n) CRel, CZ • S ↑ === S ↑ • CZ
semi-M↑CZ  : (₂₊ n) CRel, CZ • Mg ↑ === Mg ↑ • CZ^ g
semi-Ex-S↑ : (₂₊ n) CRel, Ex • S ↑ === S ↓ • Ex
semi-Ex-H↑ : (₂₊ n) CRel, Ex • H ↑ === H ↓ • Ex
blake-c12  : (₂₊ n) CRel, (S ^ p-1) ↑ • (S ^ p-1) ↓ • CX ^ p-1 • S ↓ • CX === CZ
```

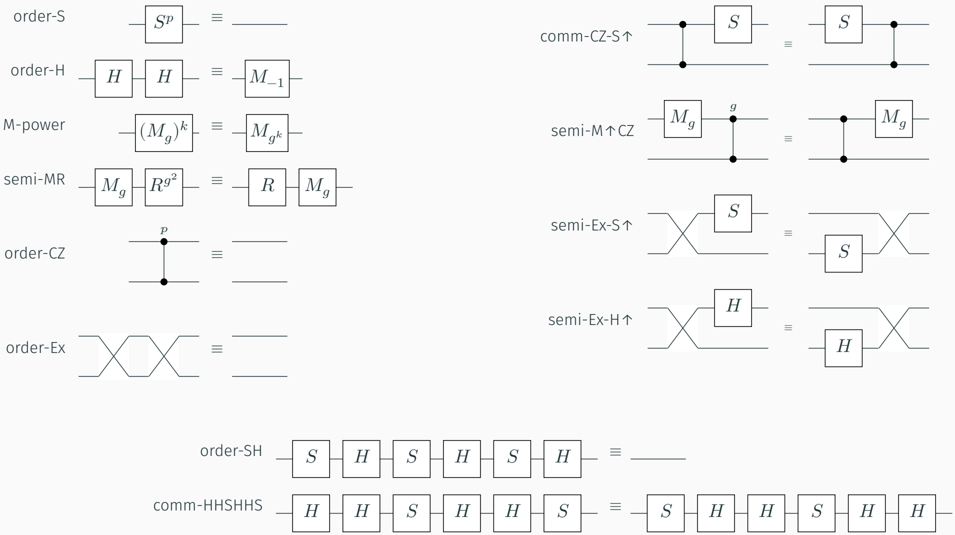
- `blake-c12` relates two circuits that implement the Controlled-Z operator.

The 16 relations, in Agda (three qudits)

```
yang-baxter :  (₃₊ n) CRel, Ex ↑ • Ex ↓ • Ex ↑ === Ex ↓ • Ex ↑ • Ex ↓
cz-slide    :  (₃₊ n) CRel, Ex ↓ • Ex ↑ • CZ === CZ ↑ • Ex ↓ • Ex ↑
semi-CX↑-CZ↓ : (₃₊ n) CRel, CZ ↓ • CX ↑ === CZ02 • CX ↑ • CZ ↓
```

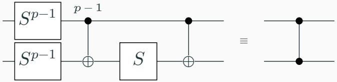
- Only these 16 are axioms. $XZ = ZX\,(\mathrm{mod}\ \omega)$, $\boldsymbol{Ex}$-vs-CZ, and both Pauli-vs-CZ rules are *derived*.
- Plus, for every gate set, the same three monoidal structural rules: congruence under $\uparrow$, gates on disjoint wires commute, and scalars commutes with every gate.

The 16 relations, as circuits (1–2 qudits)

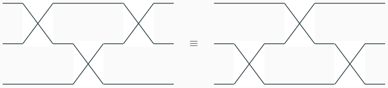


The 16 relations, as circuits (2–3 qudits)

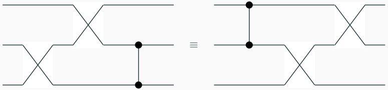
blake-c12



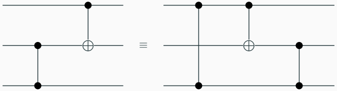
yang-baxter



cz-slide



semi-CX↑-CZ↓



What “presents” means

```
record _Presents_ (_===_ : WRel X) (G : Group a ℓ) : Set (a ⊔ ℓ) where
  field gl : Grouplike _===_
  module GL = Group-Lemmas _===_ gl
  open GroupMorphisms (Group.rawGroup GL.•-ε-group) (Group.rawGroup G)
  field
    ⟦_⟧ : Word X → Group.Carrier G
    iso : IsGroupIsomorphism ⟦_⟧
```

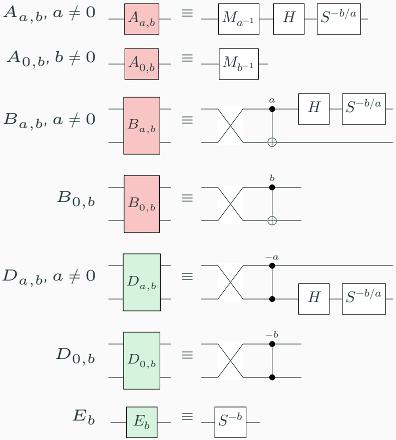
$$\text{Cir } n \,/ \approx_{\text{16 rules}} \xrightarrow[\cong]{⟦\_⟧} \mathrm{Pauli}_n \rtimes \mathrm{Sp}(2n, \mathbb{Z}_p) \;\cong\; \mathcal{C}_n/\langle\omega\rangle$$

- *Soundness*: ⟦_⟧ is well defined — each rule holds semantically.
- *Completeness*: ⟦_⟧ is injective — semantically equal circuits are related by the rules.

Main result 2: a unique normal form for symplectic operators

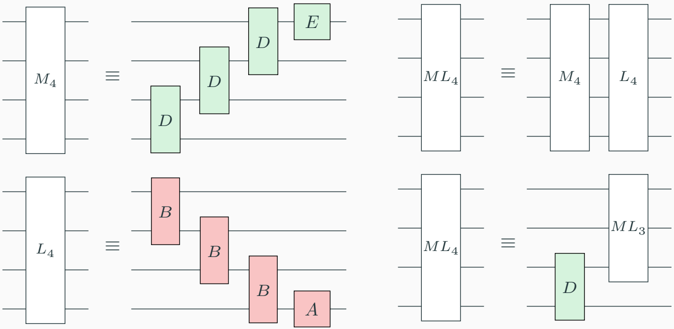
Symplectic operators are Clifford mod Pauli: the “proper” part. Normal forms are data, built from four atomic boxes:

```
A B D E : Set
A = Σ[ ab ∈ (ℤ ₚ × ℤ ₚ) ]
  (ab ≢ (₀ , ₀))
B = ℤ ₚ × ℤ ₚ
D = ℤ ₚ × ℤ ₚ
E = ℤ ₚ
```



Rows, and the staircase

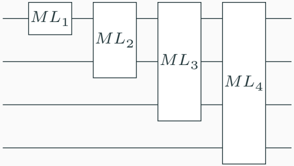
```
M L ML : ℕ → Set

M 0 = ⊤

M (1+ n) = Vec D n × E

L 0 = ⊤

L (1+ n) = Vec B n × A

ML 0 = ⊤

ML 1 = M 1 × L 1

ML m@(2+ n) =

  M m × L m ⊎ D × ML (1+ n)
```



The normal form, as a circuit

Reading normal forms back as circuits is structural recursion:

```
NF : ℕ → Set
NF 0 = ⊤
NF (1+ n) = NF n × ML (1+ n)

[_]ⁿᶠ : ∀ {n} → NF n → Word (Gen n)
[_]ⁿᶠ {0}    tt        = ε
[_]ⁿᶠ {1+ n} (nf , ml) = [ nf ]ⁿᶠ ↑ • [ ml ]ᵐˡ
```



Everything is Agda-checked

- ≈ 330 files, $\approx 74\,000$ lines of Agda, all `--safe`, no postulates.
- Claude did the semantic side code and the refactorization and improvment of the syntactic side.

What the formalisation covers

Presentations, normal forms, rewriting to normal forms, completeness.

What it does not

The two glue facts that the projective Clifford group is $\mathbf{Pauli} \rtimes \mathbf{Sp}$, and that the Clifford group is a central extension of it by $\langle \omega \rangle$, which are well-known.

Method

Divide and conquer

Factor the group; present the factors; compose the presentations.

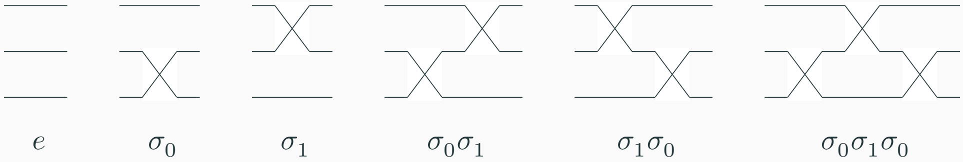
$$1 \to \mathbf{Pauli}_n \to \mathcal{C}_n/\langle\omega\rangle \to \mathbf{Sp}(2n, \mathbb{Z}_p) \to 1$$ semidirect: it splits

$$1 \to \langle\omega\rangle \to \mathcal{C}_n \to \mathcal{C}_n/\langle\omega\rangle \to 1$$ central extension

- Doesn't this just move the difficulty around? No: the symplectic factor is where the real rewriting happens, and keeping the Pauli part out of it pays off everywhere.
- Prior work (qubit: Selinger; qutrit: Li et al.) worked with the whole Clifford group at once — the derivations drag a "messy" Pauli correction through every step.

Warm-up: the symmetric group S_n

Swap circuits on n wires present S_n. All of S_3:



Circuits have a built-in tower of subgroups

$$\mathsf{Cir}\,0 \;<\; \mathsf{Cir}\,1 \;<\; \cdots \;<\; \mathsf{Cir}\,n$$

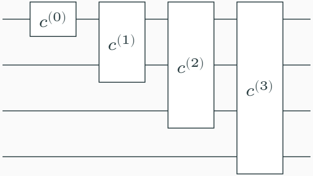
($\mathsf{Cir}\,k$ = circuits that only touch the top k wires). For S_n: $S_0 < S_1 < \cdots < S_n$.

Coset representatives, and the normal form

Representatives C_2 of S_3/S_2 (and of S_2/S_1, S_1/S_0): the staircases

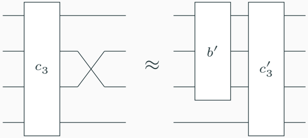
$c_0^{(2)}$ $c_1^{(2)}$ $c_2^{(2)}$ $c_0^{(1)}$ $c_1^{(1)}$ $c_0^{(0)}$

- For $f \in S_3$ there is a unique $c \in C_2$ with $c \circ f^{-1}$ fixing the bottom wire, i.e., the c that sends $f^{-1}(0)$ to 0.
- Recurse on $c \circ f^{-1} \in S_2$: every f is uniquely $c^{(0)} \cdot c^{(1)} \cdot c^{(2)}$.
- In general, given $G_0 < \cdots < G_n$ and coset reps C_k of G_k/G_{k-1}, elements of G_n factor uniquely as $\prod_k c_k$ — a mixed-radix encoding: $|S_3| = 3 \cdot 2 \cdot 1$.



Rewriting to the normal form: the coset updating table

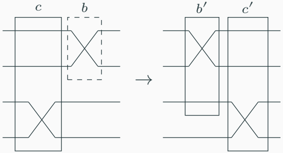
```
ract : C (₁₊ n) → Gen (₂₊ n) → Circuit (₁₊ n) × C (₁₊ n)
ract {n} ε       σ-gen       = ε , σ• ε
ract {n} (σ• ε) σ-gen       = ε , ε
ract {₁₊ n} (σ• σ• c) σ-gen = (σ {n = n}) , σ• σ• c
ract {n} ε       (g ↥)       = [ g ]ʷ , ε


ract-sound : ∀ {n} c b →
  let open PB ((₂₊ n) VRel,_===_)
      (b' , c') = ract {n} c b
  in  [ c ]ᶜ • [ b ]ʷ ≈ b' ↑ • [ c' ]ᶜ
```



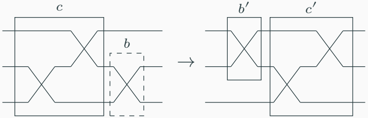
Pushing a generator b through $c_3 \in C_3$: a new coset c_3', and a residual "dirty" circuit b' on the wires above. Then we recurse on pushing b' through C_2.

Rewriting to the normal form: push a generator through

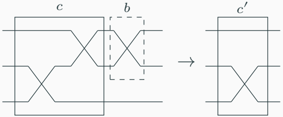
Multiply a normal form by a generator; push it through the staircase:



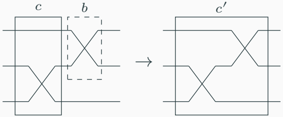
disjoint wires: commute past



Yang–Baxter: pass through, shift up



$\sigma^2 = e$: staircase shrinks



staircase grows

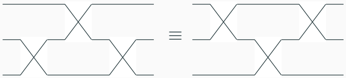
A presentation of S_n

Collecting the push-through relations and simplifying leaves just two rules (plus the structural ones):

```
data _SRel,_===_ : (n : ℕ) → WRel (Gen n) where
  order       : (₂₊ n) SRel, σ • σ === ε
  yang-baxter : (₃₊ n) SRel, σ • σ ↑ • σ === σ ↑ • σ • σ ↑
```

order

yang-baxter



```
Theorem-Sn : ∀ {n} → (n VRel,_===_) IsPresentationOf (Permutation′-group n)
Theorem-Sn = SymP.presentation
```

From S_n to $\mathrm{Sp}(2n, \mathbb{Z}_p)$: change the action

The same framework, three ways of looking at the wires:

1. S_n acts on $\{0, \ldots, n-1\}$: a wire is a *position*; c sends $f^{-1}(0)$ to 0.
2. S_n acts on **Vec Bit** n: a wire is a *bit*; $f^{-1}c$ fixes the linear subspace of the bottom wire.
3. Adding H, S, CZ: the group acts on **Vec Pauli**$_1$ n — a $2n$-dimensional symplectic $\mathbb{Z}_p$-vector space. A wire is a $\mathbf{Pauli}_1$; enlarging the coset layer from staircases to $\boldsymbol{ML}$ boxes, $f^{-1}c$ fixes the symplectic subspace of the bottom wire.

The framework applies to *any* $\mathbb{N}$-indexed family of groups with such a layered structure (`Circuit.CosetNF`, `Circuit.Uniqueness`).

Uniqueness of normal form

```
Theorem-2 : ∀ n {u v : NF n} → ⟦ [ u ] ⟧ ≈ˢ ⟦ [ v ] ⟧ → u ≡ v
Theorem-2 = U.⟦[]⟧-injective
```

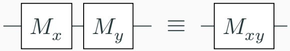
$$\begin{tikzcd}
\mathsf{Cir}\, n \,/\, \approx \arrow[rr, "⟦_⟧"] \arrow[d, bend left, "nf"] & & \mathrm{Sp}(2n, \mathbb{Z}_p) \\
NF\ n \arrow[u, bend left, "[_]"] \arrow[urr, dashed, "\text{injective}"'] & &
\end{tikzcd}$$

Soundness $+$ unique normal forms $\Rightarrow$ completeness (`by-normalization`); the interesting content is that the *coset factor is determined by the action* on Pauli states.

From 67 relations to 18

1. Rewrite *every* circuit to normal form; collect every relation the rewrite ever uses: 67 relations (box-vs-gate commutations, case by case).
2. By inspection, find a small set of primitive, bidirectional relations that derive all 67.
3. The choice is not canonical — pick the set you find natural.

An example of a derived rule, $M_x \cdot M_y = M_{xy}$:

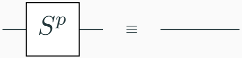



```
M-mul : ∀ n (x y : ℤ* ₚ) → let open PB (Sim._QRel,_===_ (₁₊ n)) in
        M x • M y ≈ M (x *' y)
M-mul n = SimL.Lemmas1.lemma-M-mul n
```

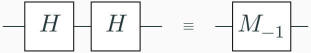
($x = g^k$, $y = g^l$; `M-power` twice; add exponents mod $p - 1$.)

The simplified symplectic rule set (1–2 qudits)

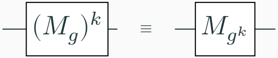
order-S



order-H



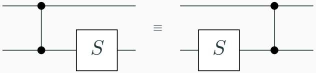
M-power



semi-MS

order-CZ

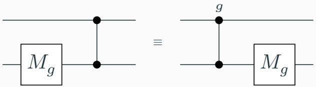
comm-CZ-S↓



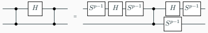
comm-CZ-S↑

semi-M↑CZ

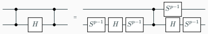
semi-M↓CZ



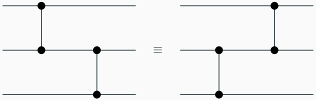
selinger-c10



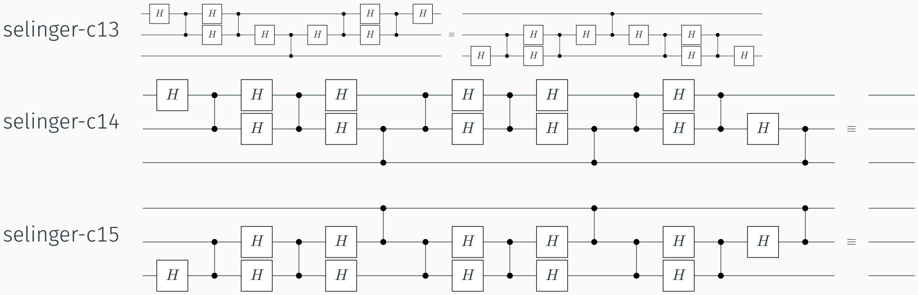
selinger-c11



selinger-c12



The simplified symplectic rule set (3 qudits), and the theorem

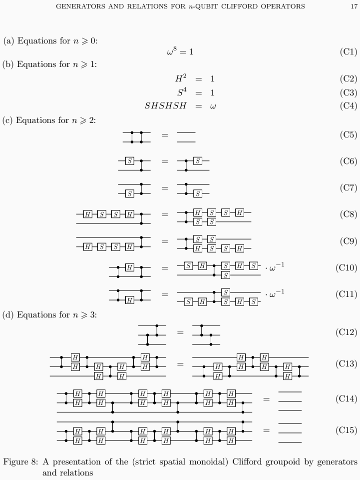



```
Theorem-Sp : ∀ {n} → (n SRel,_===_) IsPresentationOf (Sp-group n)
Theorem-Sp = SimPres.presentation
```

Comparison with the qubit rule set (Selinger)

- identical 3-qudit relations (C12–C15);
- almost identical 2-qudit relations;
- more complicated 1-qudit relations — the odd-prime multiplier calculus (M_x, primitive root g) has no qubit counterpart.

Figure 8 of Selinger, *Generators and relations for n-qubit Clifford operators* →


GENERATORS AND RELATIONS FOR n-QUBIT CLIFFORD OPERATORS 17

(a) Equations for $n \geqslant 0$:

$$\omega^8 = 1 \quad \text{(C1)}$$

(b) Equations for $n \geqslant 1$:

$$H^2 = 1 \quad \text{(C2)}$$
$$S^4 = 1 \quad \text{(C3)}$$
$$SHSHSH = \omega \quad \text{(C4)}$$

(c) Equations for $n \geqslant 2$:

(C5) (C6) (C7) (C8) (C9) (C10) (C11)

(d) Equations for $n \geqslant 3$:

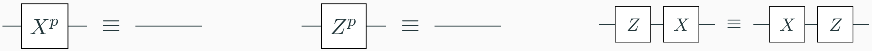
(C12) (C13) (C14) (C15)

Figure 8: A presentation of the (strict spatial monoidal) Clifford groupoid by generators and relations

Composing: the Pauli factor

The projective Pauli group $(\mathbb{Z}_p \times \mathbb{Z}_p)^n$ has an easy presentation over generators X, Z on each wire:

```
data _PRel,_===_ : (n : ℕ) → WRel (Gen n) where
  order-X  : (₁₊ n) PRel, X ^ p === ε
  order-Z  : (₁₊ n) PRel, Z ^ p === ε
  comm-Z-X : (₁₊ n) PRel, Z • X === X • Z

Theorem-Pauli : ∀ {n} → (XZ._QRel,_===_ n) IsPresentationOf (+ₚ-group n)
Theorem-Pauli = XZ.presentation
```



Composing: the semidirect product

Semidirect product: relations of both factors, plus twisted commuting, $h\,n' = ({}^{h}n')\,h$, so $(n,h)\cdot(n',h') = (n\cdot{}^{h}n'\,,\;h\,h')$:

```
conj : Sym.Gen n → XZ.Gen n → Word (XZ.Gen n)
conj Sym.H-gen  XZ.X-gen         = XZ.Z
conj Sym.H-gen  XZ.Z-gen         = XZ.X ^ p-1
conj Sym.S-gen  XZ.X-gen         = XZ.X • XZ.Z
conj Sym.S-gen  XZ.Z-gen         = XZ.Z
conj Sym.CZ-gen XZ.X-gen         = XZ.X • XZ.Z XZ.↑
conj Sym.CZ-gen (XZ.X-gen XZ.↥) = XZ.X XZ.↑ • XZ.Z

Pauli⋊Symplectic : (n : ℕ) → WRel (XZ.Gen n ⊎ Sym.Gen n)
Pauli⋊Symplectic n = XZ._QRel,_===_ n ⋄ Sim._QRel,_===_ n ⋄ ConjRelʷ conj
```

The projection to the symplectic factor is a homomorphism, so symplectic circuits

Plumbing: presentation simplification

Two presentations are equivalent when the presented groups are isomorphic over the same interpretation — each side's axioms derived from the other's. All Agda-checked, e.g. `Paper-V0.Iso`, `Simplified.Iso`.

	`SemiDirect`	`Paper-V0`
generators	X, Z, H, S, CZ per wire	H, S, CZ per wire
relations	Pauli + symplectic + conjugation	16

generator reduction: X, Z become the derived words from the introduction

relation reduction: conjugation rules become derivable, e.g. $CZ \cdot X\!\uparrow \approx X\!\uparrow \cdot Z\!\downarrow \cdot CZ$

Getting the scalars back

The Clifford group itself is a central extension by $\langle\omega\rangle$. On presentations: add a generator ω with $\omega^p = \varepsilon$, make it commute with every gate, and correct each mod-scalar relation by the scalar it actually picks up — only `order-SH` needs one:

$$-\boxed{S}-\boxed{H}-\boxed{S}-\boxed{H}-\boxed{S}-\boxed{H}- \;\equiv\; \boxed{\omega^{(p^2-1)/8}\, I}$$

```
Theorem-Clifford : ∀ (n : ℕ) →
  (QS._Exact,_===_ n) IsPresentationOf
  (PE.Presented-group n (CQP.exact-generator-data n))
Theorem-Clifford = CQP.presentation-exact
```

(δ_p : the exponent $(p^2-1)/8$ is exact division — p odd — and the correction is checked, not assumed.)

Conclusion

Conclusion

- Main result 1: presentations by generators and relations of the qudit Clifford operators (16 relations, ≤ 3 qudits) and of the symplectic operators, for all odd primes at once.
- Main result 2: a unique normal form for qudit Clifford and symplectic operators.
- Both are checked in Agda — except for the two classical glue facts: that projective Clifford is $\mathbf{Pauli} \rtimes \mathbf{Sp}$, and that Clifford is a central extension of projective Clifford by the scalars.

Thank you!